

◦ Aim:

- To estimate the concentration of the given strong acid, HCl by conductometric titration with a strong base, NaOH.

◦ Principle:

- In conductometric titrations, the conductance of the solution is measured during the titration, before and after the equivalence point, and the variance of conductance of the solution during the titration is used to locate the endpoint.
- The strong acid HCl undergoes almost complete dissociation in solution and the conductance of the solution is due to H^+ and Cl^- ions. When the strong base, NaOH (which exists as Na^+ and OH^- in the solution) is added, it neutralizes HCl to NaCl and weakly dissociating water. NaCl being a strong electrolyte remains in the solution as Na^+ and Cl^- ions.
$$H^+ Cl^- + Na^+ OH^- \rightarrow Na^+ Cl^- + H_2O$$
- During the titration, the net change is highly conducting H^+ ions in the solution are substituted by less mobile Na^+ ions. Therefore, there is a decrease in conductance as more and more NaOH is added to neutralize HCl. This continues till all the acid is neutralized by the base. After the endpoint, the addition of NaOH remains as excess, furnishing more Na^+ and OH^- ions. As the quantity of NaOH increases, the concentration of Na^+ and OH^- ions increases, thereby increasing the conductance of the solution. Therefore, during the titration, the conductance decreases sharply till the endpoint and then increases. The conductance of the solution is plotted against volume of the base added and the equivalence point corresponds to the point of

° Observations:

Vol. of NaOH (ml)	Conductance	Vol. of NaOH (ml)	Conductance
0	11.20	5.0	4.54
0.5	10.27	5.5	4.00
1.0	9.54	6.0	3.44
1.5	8.84	6.5	2.99
2.0	8.14	7.0	2.66
2.5	7.46	7.5	2.74
3.0	6.81	8.0	2.84
3.5	6.13	8.5	2.98
4.0	5.69	9.0	3.12
4.5	5.10	9.5	3.38

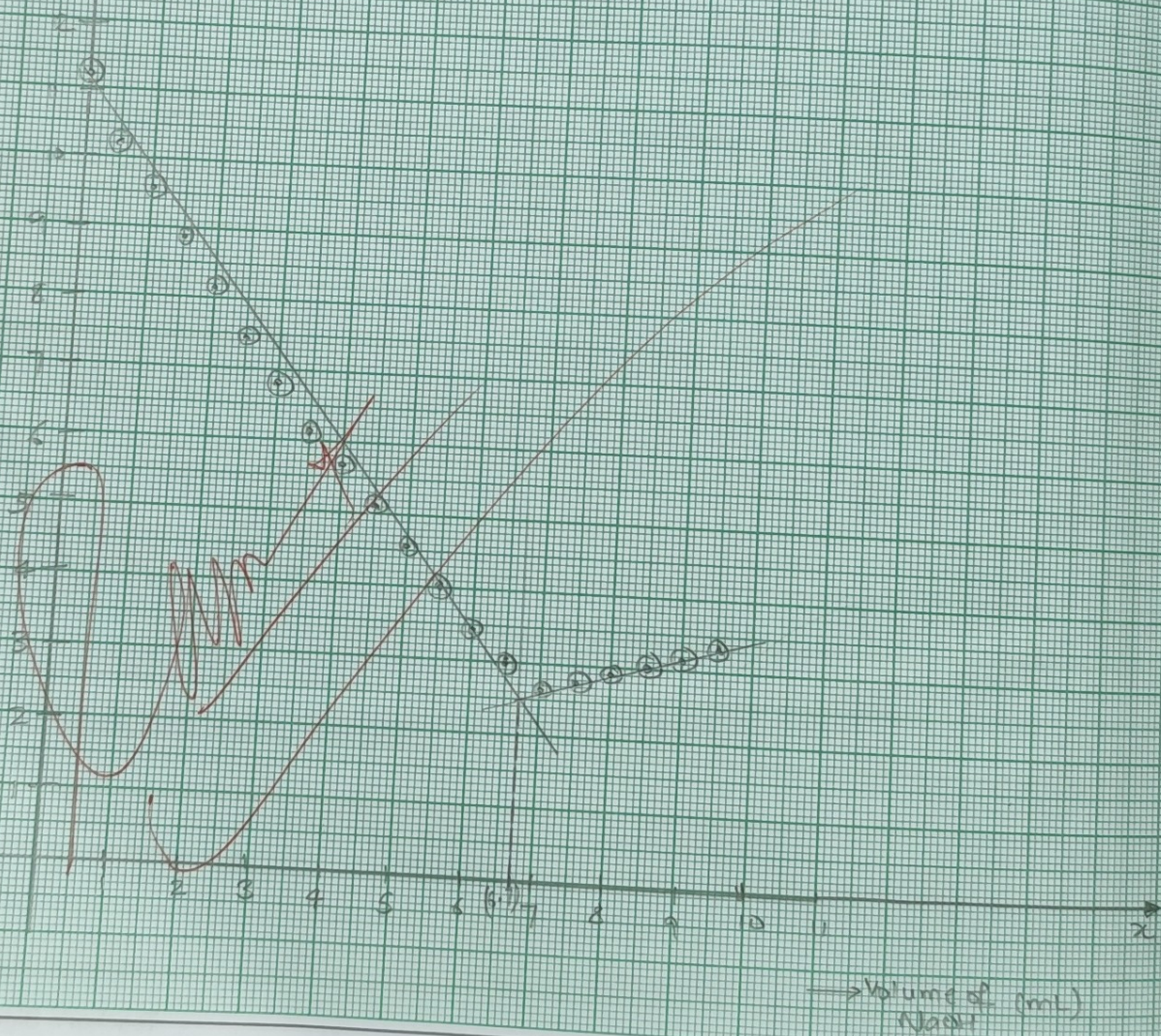
° Calculations:

- By graph, at endpoint, Volume of NaOH (V_{NaOH}) = 6.7 mL
- Concentration of NaOH (N_{NaOH}) = 0.1M
- Volume of HCl (V_{HCl}) = 25 mL
- ∴ Concentration of HCl,

$$N_{HCl} = \frac{N_{NaOH} \cdot V_{NaOH}}{V_{HCl}} = \frac{(0.1)(6.7)}{25} = \underline{\underline{0.0268M}}$$

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States: 2017: 0.15, 1990-1991: 0.17, 2018: 0.16, 1990-1991: 0.15



intersection of two straight line portions. The solution taken in the burette should be at least ten times more concentrated than the analyte solution to avoid effect of dilution on variance of conductance.

• Procedure:

- Pipette out 25 mL of HCl into a beaker.
- Clean the conductance cell with distilled water and dip it in the acid.
- Connect the conductivity cell to the conductometer and select a suitable range for conductance.
- Fill the burette with a strong base, NaOH.
- Add 0.5 mL of the base to the beaker at a time, stir well and note the conductance of the solution after each addition.
- The titration should be continued till the decreasing trend of conductance changes to an increasing trend and take 5-6 more readings.
- Plot the Conductance values (y-axis) against volume of NaOH (x-axis) and find the point of intersection of the two straight lines.

• Result:

- conductance of given strong acid = 0.0268 M